

Power Strip Model Rubric

How your P.1 Lesson 2 Power Strip Model assignment is graded

OVERVIEW

This page shows how your Power Strip Model assignment is graded. Part A asks you to list the key structures of the power strip and tell what each one does. Part B asks you to draw a model that shows how electrical energy flows through the power strip to power at least two devices. Each part is graded with five levels. The levels describe how complete and clear your reasoning is. There is no single required list of parts and no single required drawing.

How this assignment is graded

The assignment has two parts. Each part is graded on one skill, with five levels for each skill.

1. **Part A: key structures.** In Part A you list the parts of the power strip and tell what each part does. Skill 1 is graded on how completely you name parts and how clearly you explain their purposes.
2. **Part B: energy-flow model.** In Part B you draw a labeled model showing how energy travels from the source through the strip to at least two devices. Skill 2 is graded on how complete the path, the labels, and the flow direction are.
3. **Levels show depth, not one right answer.** A higher level means your reasoning is more complete and more connected. Different students can reach the top level with different parts and different drawings.

Question 1: Part A, Key Structures

Skill: I am able to identify key components in the power strip and recognize their purpose.

Missing 0%

Missing: There is no attempt to identify parts of the power strip or their purpose.

Emerging 55%

Emerging: There is an attempt to identify parts of the power strip

- BUT -

the parts named are mostly outside the strip (only the wall outlet, the cord, or the devices are mentioned)

- OR -

the parts are named without any sense of what they do.

For example:

- "the plug and the cord"
- "the outlets where you plug things in"

Approaching 65%

Approaching: There is a partial attempt to identify parts of the power strip

-BECAUSE-

at least one part inside the strip is named

-BUT-

the purpose of most parts is missing, vague, or restated as "it gives power" without explaining how.

For example:

- "the metal pieces inside that carry the electricity"
- "the switch turns it on and off"

Developing 85%

Developing: There is a solid attempt to identify parts of the power strip

-BECAUSE-

parts inside the strip are named along with parts you can see from outside

-AND-

a purpose is given for most of the named parts

-BUT-

the purposes may be loosely worded, or the list leaves out a part that matters for sending energy to more than one device.

Refining 100%

Refining: There is a strong attempt to identify parts of the power strip

-BECAUSE-

visible parts and internal parts are both named with a clear purpose for each

-AND-

the purposes add up to a story of how energy can be sent from one source to several devices at once.

Question 2: Part B, Model Diagram

Skill: I am able to create a model which shows how the electrical energy flows within the power strip in order to use the energy source to power at least two devices.

Missing 0%

Missing: There is no attempt to draw a model of the power strip.

Emerging 55%

Emerging: There is an attempt to draw a model

- BUT -

the drawing shows only the outside of the power strip or only one device

- OR -

there is no visible path for energy to travel from the source to any device.

For example:

- a rectangle labeled "power strip" with one phone plugged in and no arrows
- a drawing of a wall outlet only

Approaching 65%

Approaching: There is a partial attempt to draw a model

-BECAUSE-

the drawing shows the source and at least one device

-AND-

some labels are present

-BUT-

the energy path does not reach every device shown,

-OR-

the labels name objects without showing how they connect.

For example:

- a source and two devices drawn but only one connected line
- labels like "cord" and "lamp" with no arrows

Developing 85%

Developing: There is a solid attempt to draw a model

-BECAUSE-

the drawing shows one energy source connected to the power strip

-AND-

the path from the strip reaches at least two devices

-AND-

most key parts are labeled

-BUT-

the flow direction is missing or unclear at some points,

-OR-

one important internal part is unlabeled or missing.

Refining 100%

Refining: There is a strong attempt to draw a model

-BECAUSE-

the drawing shows one energy source flowing through the power strip

-AND-

the path clearly branches to power at least two devices

-AND-

visible and internal parts are labeled

-AND-

arrows or another clear device shows the direction the energy travels.

Tier ladder

Missing · 0%

No attempt at this part.

Emerging · 55%

An attempt is there, but the skill is just starting.

Approaching · 65%

The skill is partly there, with important pieces missing.

Developing · 85%

The skill is mostly there, with a small gap left.

Refining · 100%

The skill is clearly there, with complete and connected reasoning.